

Notebook LLM Clone

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Problem

Students face difficulty while preparing for exams. They may have PDFs, but those documents are often very long, making it difficult to revise quickly and understand concepts in a short amount of time.

Solution

Build a RAG (Retrieval-Augmented Generation) system where students can upload different types of study materials such as:

- PDFs
- Images
- Text files
- DOC files
- Other document formats

The AI model processes these files and provides answers / features such as:

- Summarization
- Question & Answer (Q&A)
- Concept Explanation
- (Future add-on) Search the internet for additional information

Clarifying Requirements

1. What does the Knowledge Base (KB) consist of?

- PDFs
- DOC files
- Images

2. Does the data change over time?

Answer: **No**.

3. Does the KB contain only text?

Answer: **No** — it may contain:

- Text
- Diagrams
- Charts
- Images
- Tables
- etc.

4. How many pages? What is the expected growth?

- 6 subjects
- Each subject has 2 PDFs
- Each PDF has around 100 pages
- More than 500 students

Estimated pages: $500 \times 6 \times 2 = 6000+$ pages. Future growth needs to be estimated as well.

5. Is metadata required?

Answer: **Yes**. Metadata generated from uploaded documents should include:

- Page number
- Chapter
- etc.

6. Should the system support multiple languages?

Answer: **No** — support only English for now.

7. Does the system need to respond in real time?

Answer: **Not necessarily** (to be confirmed).

8. Do we need to address safety concerns, harmful outputs, or misleading outputs?

Answer: **Not right now**.

Technical Decisions

1. Document Parsing

Use **AI-based document parsing**, because the documents contain complex structures such as:

- Tables
- Images
- Mixed layouts
- Other complex data

2. Document Chunking

Chunking is very important because retrieval quality depends on it.

Goal: retrieve the most meaningful and relevant sections of a document.

Possible chunking methods:

- Fixed chunking
- Regular-expression-based chunking
- HTML-based chunking
- Semantic chunking ✓ **(preferred)**

Semantic chunking reduces the need for manually writing parsing rules.

3. Embedding Model Choice

Recommended: **text-embedding-3-small**.

Reason: suitable for this data size, with a good balance between cost and performance.

Suitable AI Approach

Fine-tuning ✗ **Not recommended**

- Responses are not highly domain-specific.
- Computationally expensive.
- Requires retraining whenever data changes.
- More maintenance overhead.
- Providing references / citations becomes difficult.
- Students cannot verify the original source easily.

Prompt Engineering only ✗ **Not sufficient**

- Relies only on the model's built-in knowledge.
- Cannot effectively use the uploaded documents.

RAG (Retrieval-Augmented Generation) ✓ Recommended

- Uses the uploaded study materials.
- Can provide document-grounded answers.
- Easier to maintain.
- Lower cost than fine-tuning.
- Can provide references to source documents.

Retrieval Strategy

Keyword-based retrieval ✗ Not preferred

Knowledge-graph-based retrieval ✗ Not needed

- No entity-relationship or hierarchical matching is required.

Vector-based retrieval ✓ Recommended

- Uses embeddings to find semantically similar content.

Reranking

Current decision: **not needed for now.**